


```

7   generated_outputs_lora = []
8   for text in pipeline_iterator:
9       generated_outputs_lora.append(text[0]["generated_text"])

```

mistral-8x7b-instruct-v01 watsonx.ai inference model object

Adjusts the parameters to push the limits of creativity and response length.

Code Example:

```

1   model_id = 'mistralai/mistral-8x7b-instruct-v01'
2   parameters = {
3       GenParams.MAX_NEW_TOKENS: 256, # this controls the maximum number of tokens in the generated response
4       GenParams.TEMPERATURE: 0.5, # this randomness or creativity of the model's responses
5   }
6   credentials = {
7       "url": "https://us-south.ml.cloud.ibm.com"
8   }
9   project_id = "skills-network"
10  model = ModelInference(
11      model_id=model_id,
12      params=parameters,
13      credentials=credentials,
14      project_id=project_id
15  )

```

String prompt templates

Used to format a single string and are generally used for simpler inputs.

Code Example:

```

1   from langchain_core.prompts import PromptTemplate
2   prompt = PromptTemplate.from_template("Tell me one {adjective} joke about {topic}")
3   input_ = {"adjective": "funny", "topic": "cats"} # create a dictionary to store the corresponding values
4   prompt.invoke(input_)

```

Chat prompt templates

Used to format a list of messages. These "templates" consist of a list of templates themselves.

Code Example:

```

1   from langchain_core.prompts import ChatPromptTemplate
2   prompt = ChatPromptTemplate.from_messages([
3       ("system", "You are a helpful assistant"),
4       ("user", "Tell me a joke about {topic}")
5   ])
6   input_ = {"topic": "cats"}
7   prompt.invoke(input_)

```

Messages place holder

This prompt template is responsible for adding a list of messages in a particular place. But if you want the user to pass in a list of messages that you would slot into a particular spot, the given code snippet is helpful.

Code Example:

```

1   from langchain_core.prompts import MessagesPlaceholder
2   from langchain_core.messages import HumanMessage
3   prompt = ChatPromptTemplate.from_messages([
4       ("system", "You are a helpful assistant"),
5       MessagesPlaceholder("msgs")
6   ])
7   input_ = {"msgs": [HumanMessage(content="What is the day after Tuesday?")]}
8   prompt.invoke(input_)

```

Example selector

If you have many examples, you may need to select which ones to include in the prompt. The Example Selector is the class responsible for doing so.

Code Example:

```

1   from langchain_core.example_selectors import LengthBasedExampleSelector

```

```

2 from langchain_core.prompts import FewShotPromptTemplate, PromptTemplate
3 # Examples of a pretend task of creating antonyms.
4 examples = [
5     {"input": "happy", "output": "sad"},
6     {"input": "tall", "output": "short"},
7     {"input": "energetic", "output": "lethargic"},
8     {"input": "sunny", "output": "gloomy"},
9     {"input": "windy", "output": "calm"},
10 ]
11 example_prompt = PromptTemplate(
12     input_variables=["input", "output"],
13     template="Input: {input}\nOutput: {output}",
14 )
15 example_selector = LengthBasedExampleSelector(
16     examples=examples,
17     example_prompt=example_prompt,
18     max_length=25, # The maximum length that the formatted examples should be.
19 )
20 dynamic_prompt = FewShotPromptTemplate(
21     example_selector=example_selector,
22     example_prompt=example_prompt,
23     prefix="Give the antonym of every input",
24     suffix="Input: {adjective}\nOutput:",
25     input_variables=["adjective"],
26 )

```

JSON parser

This output parser allows users to specify an arbitrary JSON schema and query LLMs for outputs that conform to that schema.

Code Example:

```

1 from langchain_core.output_parsers import JsonOutputParser
2 from langchain_core.pydantic_v1 import BaseModel, Field
3 # Define your desired data structure.
4 class Joke(BaseModel):
5     setup: str = Field(description="question to set up a joke")
6     punchline: str = Field(description="answer to resolve the joke")
7 # And a query intended to prompt a language model to populate the data structure.
8 joke_query = "Tell me a joke."
9 # Set up a parser + inject instructions into the prompt template.
10 output_parser = JsonOutputParser(pydantic_object=Joke)
11 format_instructions = output_parser.get_format_instructions()
12 prompt = PromptTemplate(
13     template="Answer the user query.\n{format_instructions}\n{query}\n",
14     input_variables=["query"],
15     partial_variables={"format_instructions": format_instructions},
16 )
17 chain = prompt | mixtral_llm | output_parser
18 chain.invoke({"query": joke_query})

```

Comma separated list parser

This output parser can be used when you want to return a list of comma-separated items.

Code Example:

```

1 from langchain.output_parsers import CommaSeparatedListOutputParser
2 output_parser = CommaSeparatedListOutputParser()
3 format_instructions = output_parser.get_format_instructions()
4 prompt = PromptTemplate(
5     template="Answer the user query. {format_instructions}\nList five {subject}.",
6     input_variables=["subject"],
7     partial_variables={"format_instructions": format_instructions},
8 )
9 chain = prompt | mixtral_llm | output_parser

```

Document object

Contains information about some data in LangChain. It has two attributes: `page_content`:

- **str**: This attribute holds the content of the document.
- **metadata: dict**: This attribute contains arbitrary metadata associated with the document. It can be used to track various details such as the document id, file name, and so on.

Code Example:

```

1 from langchain_core.documents import Document
2 Document(page_content="Python is an interpreted high-level general-purpose programming language. Python's design philosophy emphasizes code readability with its notable syntax. Python's design philosophy emphasizes code readability with its notable syntax.",
3         metadata={
4

```

```

5         'my_document_id' : 234234,
6         'my_document_source' : "About Python",
7         'my_document_create_time' : 1680013019
8     })

```

text_splitter

At a high level, text splitters work as follows:

- Split the text into small, semantically meaningful chunks (often sentences).
- Start combining these small chunks into a larger chunk until you reach a certain size (as measured by some function).
- Once you reach that size, make that chunk its own piece of text and start creating a new chunk with some overlap (to keep context between chunks).

Code Example:

```

1 text_splitter = CharacterTextSplitter(chunk_size=200, chunk_overlap=20, separator="\n") # <
2 chunks = text_splitter.split_documents(document)
3 print(len(chunks))

```

Embedding models

Embedding models are specifically designed to interface with text embeddings. Embeddings generate a vector representation for a given piece of text. This is advantageous as it allows you to conceptualize text within a vector space. Consequently, you can perform operations such as semantic search, where you identify pieces of text that are most similar within the vector space.

Code Example:

```

1 from ibm_watsonx_ai.metanames import EmbedTextParamsMetaNames
2 embed_params = {
3     EmbedTextParamsMetaNames.TRUNCATE_INPUT_TOKENS: 3,
4     EmbedTextParamsMetaNames.RETURN_OPTIONS: {"input_text": True},
5 }
6 from langchain_ibm import WatsonxEmbeddings
7 watsonx_embedding = WatsonxEmbeddings(
8     model_id="ibm/slate-125m-english-rtrvr",
9     url="https://us-south.ml.cloud.ibm.com",
10    project_id="skills-network",
11    params=embed_params,
12 )

```

Vector store-backed retriever

A retriever that uses a vector store to retrieve documents. It is a lightweight wrapper around the vector store class to make it conform to the retriever interface. It uses the search methods implemented by a vector store, like similarity search and MMR (maximum marginal relevance), to query the texts in the vector store. Since we've constructed a vector store docsearch, it's very easy to construct a retriever.

Code Example:

```

1 retriever = docsearch.as_retriever()
2 docs = retriever.invoke("Langchain")A vector store retriever is a retriever that uses a vect
3 Since we've constructed a vector store docsearch, it's very easy to construct a retriever.

```

ChatMessageHistory class

One of the core utility classes underpinning most (if not all) memory modules is the ChatMessageHistory class. This super lightweight wrapper provides convenient methods for saving HumanMessages, AIMessages, and then fetching them all.

Code Example:

```

1 from langchain.memory import ChatMessageHistory
2 chat = mixtral_llm
3 history = ChatMessageHistory()
4 history.add_ai_message("hi!")
5 history.add_user_message("what is the capital of France?")

```


langchain.chains

This code snippet uses a LangChain, library for building language model applications, creating a chain to generate popular dish recommendations based on the specified locations. It also configures model inference settings for further processing.

Code Example:

```
1 from langchain.chains import LLMChain
2 template = """Your job is to come up with a classic dish from the area that the users suggest
3 {location}
4 YOUR RESPONSE:
5 """
6 prompt_template = PromptTemplate(template=template, input_variables=['location'])
7 # chain 1
8 location_chain = LLMChain(llm=mixtral_llm, prompt=prompt_template, output_key='meal')
9 location_chain.invoke(input={'location':'China'})
```

Simple sequential chain

Sequential chains allow the output of one LLM to be used as the input for another. This approach is beneficial for dividing tasks and maintaining the focus of your LLM.

Code Example:

```
1 from langchain.chains import SequentialChain
2 template = """Given a meal {meal}, give a short and simple recipe on how to make that dish
3 YOUR RESPONSE:
4 """
5 prompt_template = PromptTemplate(template=template, input_variables=['meal'])
6 # chain 2
7 dish_chain = LLMChain(llm=mixtral_llm, prompt=prompt_template, output_key='recipe')
8 template = """Given the recipe {recipe}, estimate how much time I need to cook it.
9 YOUR RESPONSE:
10 """
11 prompt_template = PromptTemplate(template=template, input_variables=['recipe'])
12 # chain 3
13 recipe_chain = LLMChain(llm=mixtral_llm, prompt=prompt_template, output_key='time')
14 # overall chain
15 overall_chain = SequentialChain(chains=[location_chain, dish_chain, recipe_chain],
16 input_variables=['location'],
17 output_variables=['meal', 'recipe', 'time'],
18 verbose= True)
```

load_summarize_chain

This code snippet uses LangChain library for loading and using a summarization chain with a specific language model and chain type. This chain type will be applied to web data to print a resulting summary.

Code Example:

```
1 from langchain.chains.summarize import load_summarize_chain
2 chain = load_summarize_chain(llm=mixtral_llm, chain_type="stuff", verbose=False)
3 response = chain.invoke(web_data)
4 print(response['output_text'])n
```

TextClassifier

Represents a simple text classifier that uses an embedding layer, a hidden linear layer with a ReLU activation, and an output linear layer. The constructor takes the following arguments:

- **num_class:** The number of classes to classify.
- **freeze:** Whether to freeze the embedding layer.

Code Example:

```
1 from torch import nn
2 class TextClassifier(nn.Module):
3     def __init__(self, num_classes, freeze=False):
4         super(TextClassifier, self).__init__()
5         self.embedding = nn.Embedding.from_pretrained(glove_embedding.vectors.to(device), freeze=freeze)
6         # An example of adding additional layers: A linear layer and a ReLU activation
7         self.fc1 = nn.Linear(in_features=100, out_features=128)
8         self.relu = nn.ReLU()
9         # The output layer that gives the final probabilities for the classes
10        self.fc2 = nn.Linear(in_features=128, out_features=num_classes)
11    def forward(self, x):
12        # Pass the input through the embedding layer
13        x = self.embedding(x)
```

```

13         x = self.embedding(x)
14         # Here you can use a simple mean pooling
15         x = torch.mean(x, dim=1)
16         # Pass the pooled embeddings through the additional layers
17         x = self.fc1(x)
18         x = self.relu(x)
19         return self.fc2(x)

```

Train the model

This code snippet outlines the function to train a machine learning model using PyTorch. This function trains the model over a specified number of epochs, tracks them, and evaluates the performance on the data set.

Code Example:

```

1  def train_model(model, optimizer, criterion, train_dataloader, valid_dataloader, epochs=100):
2      cum_loss_list = []
3      acc_epoch = []
4      best_acc = 0
5      file_name = model_name
6      for epoch in tqdm(range(1, epochs + 1)):
7          model.train()
8          cum_loss = 0
9          for _, (label, text) in enumerate(train_dataloader):
10             optimizer.zero_grad()
11             predicted_label = model(text)
12             loss = criterion(predicted_label, label)
13             loss.backward()
14             torch.nn.utils.clip_grad_norm_(model.parameters(), 0.1)
15             optimizer.step()
16             cum_loss += loss.item()
17             #print("Loss:", cum_loss)
18             cum_loss_list.append(cum_loss)
19             acc_val = evaluate(valid_dataloader, model, device)
20             acc_epoch.append(acc_val)
21             if acc_val > best_acc:
22                 best_acc = acc_val
23                 print(f"New best accuracy: {acc_val:.4f}")
24                 #torch.save(model.state_dict(), f"{model_name}.pth")
25             #save_list_to_file(cum_loss_list, f"{model_name}_loss.pkl")
26             #save_list_to_file(acc_epoch, f"{model_name}_acc.pkl")

```

llm_model

This code snippet defines function 'llm_model' for generating text using the language model from the mistral.ai platform, specifically the 'mistral-8x7b-instruct-v01' model. The function helps in customizing generating parameters and interacts with IBM Watson's machine learning services.

Code Example:

```

1  def llm_model(prompt_txt, params=None):
2      model_id = 'mistralai/mistral-8x7b-instruct-v01'
3      default_params = {
4          "max_new_tokens": 256,
5          "min_new_tokens": 0,
6          "temperature": 0.5,
7          "top_p": 0.2,
8          "top_k": 1
9      }
10     if params:
11         default_params.update(params)
12     parameters = {
13         GenParams.MAX_NEW_TOKENS: default_params["max_new_tokens"], # this controls the max
14         GenParams.MIN_NEW_TOKENS: default_params["min_new_tokens"], # this controls the min
15         GenParams.TEMPERATURE: default_params["temperature"], # this randomness or creativity
16         GenParams.TOP_P: default_params["top_p"],
17         GenParams.TOP_K: default_params["top_k"]
18     }
19     credentials = {
20         "url": "https://us-south.ml.cloud.ibm.com"
21     }
22     project_id = "skills-network"
23     model = Model(
24         model_id=model_id,
25         params=parameters,
26         credentials=credentials,
27         project_id=project_id
28     )
29     mistral_llm = WatsonxLLM(model=model)
30     response = mistral_llm.invoke(prompt_txt)
31     return response

```

Zero-shot prompt

Zero-shot learning is crucial for testing a model's ability to apply its pre-trained knowledge to new, unseen tasks without additional training. This capability is valuable for gauging the model's generalization skills.

Code Example:

```
1 prompt = """Classify the following statement as true or false:
2         'The Eiffel Tower is located in Berlin.'
3         Answer:
4     """
5 response = llm_model(prompt, params)
6 print(f"prompt: {prompt}\n")
7 print(f"response : {response}\n")
```

One-shot prompt

One-shot learning example where the model is given a single example to help guide its translation from English to French. The prompt provides a sample translation pairing, "How is the weather today?" translated to "Comment est le temps aujourd'hui?" This example serves as a guide for the model to understand the task context and desired format. The model is then tasked with translating a new sentence, "Where is the nearest supermarket?" without further guidance.

Code Example:

```
1 params = {
2     "max_new_tokens": 20,
3     "temperature": 0.1,
4 }
5 prompt = """Here is an example of translating a sentence from English to French:
6         English: "How is the weather today?"
7         French: "Comment est le temps aujourd'hui?"
8         Now, translate the following sentence from English to French:
9         English: "Where is the nearest supermarket?"
10    """
11 response = llm_model(prompt, params)
12 print(f"prompt: {prompt}\n")
13 print(f"response : {response}\n")
```

Few-shot prompt

This code snippet classifies emotions using a few-shot learning approach. The prompt includes various examples where statements are associated with their respective emotions.

Code Example:

```
1 #parameters `max_new_tokens` to 10, which constrains the model to generate brief responses
2 params = {
3     "max_new_tokens": 10,
4 }
5 prompt = """Here are few examples of classifying emotions in statements:
6         Statement: 'I just won my first marathon!'
7         Emotion: Joy
8         Statement: 'I can't believe I lost my keys again.'
9         Emotion: Frustration
10        Statement: 'My best friend is moving to another country.'
11        Emotion: Sadness
12        Now, classify the emotion in the following statement:
13        Statement: 'That movie was so scary I had to cover my eyes.'
14    """
15 response = llm_model(prompt, params)
16 print(f"prompt: {prompt}\n")
17 print(f"response : {response}\n")
```

Chain-of-thought (CoT) prompting

The Chain-of-Thought (CoT) prompting technique, designed to guide the model through a sequence of reasoning steps to solve a problem.

The CoT technique involves structuring the prompt by instructing the model to "Break down each step of your calculation." This encourages the model to include explicit reasoning steps, mimicking human-like problem-solving processes.

Code Example:

```
1 params = {
2     "max_new_tokens": 512,
3     "temperature": 0.5,
4 }
5 prompt = """Consider the problem: 'A store had 22 apples. They sold 15 apples today and got
6         How many apples are there now?'
7         Break down each step of your calculation
```

```

7         # Break down each step of your calculation
8         """
9         response = llm_model(prompt, params)
10        print(f"prompt: {prompt}\n")
11        print(f"response : {response}\n")

```

Self-consistency

This code snippet determines the consistent result for age-related problems and generates multiple responses. The 'params' dictionary specifies the maximum number of tokens to generate responses.

Code Example:

```

1  params = {
2      "max_new_tokens": 512,
3  }
4  prompt = """When I was 6, my sister was half of my age. Now I am 70, what age is my sister?
5          Provide three independent calculations and explanations, then determine the most
6  """
7  response = llm_model(prompt, params)
8  print(f"prompt: {prompt}\n")
9  print(f"response : {response}\n")

```

Prompt template

A key concept in LangChain, it helps to translate user input and parameters into instructions for a language model. This can be used to guide a model's response, helping it understand the context and generate relevant and coherent language-based output.

Code Example:

```

1  model_id = 'mistralai/mixtral-8x7b-instruct-v01'
2  parameters = {
3      GenParams.MAX_NEW_TOKENS: 256, # this controls the maximum number of tokens in the generated response
4      GenParams.TEMPERATURE: 0.5, # this randomness or creativity of the model's responses
5  }
6  credentials = {
7      "url": "https://us-south.ml.cloud.ibm.com"
8  }
9  project_id = "skills-network"
10 model = Model(
11     model_id=model_id,
12     params=parameters,
13     credentials=credentials,
14     project_id=project_id
15 )
16 mixtral_llm = WatsonxLLM(model=model)
17 mixtral_llm

```

Text summarization

Text summarization agent designed to help summarize the content you provide to the LLM. You can store the content to be summarized in a variable, allowing for repeated use of the prompt.

Code Example:

```

1  content = """
2      The rapid advancement of technology in the 21st century has transformed various industries.
3      Innovations such as artificial intelligence, machine learning, and the Internet of Things (IoT) have
4      For instance, AI-powered diagnostic tools are improving the accuracy and speed of medical diagnoses.
5      Moreover, online learning platforms are making education more accessible to people across the globe.
6      These technological developments are not only enhancing productivity but also contributing to economic growth.
7  """
8  template = """Summarize the {content} in one sentence.
9  """
10 prompt = PromptTemplate.from_template(template)
11 llm_chain = LLMChain(prompt=prompt, llm=mixtral_llm)
12 response = llm_chain.invoke(input = {"content": content})
13 print(response["text"])

```

Question answering

An agent that enables the LLM to learn from the provided content and answer questions based on what it has learned. Occasionally, if the LLM does not have sufficient information, it might generate a speculative answer. To manage this, you'll specifically instruct it to respond with "Unsure about the answer" if it is uncertain about the correct response.

Code Example:

```

1 content = """
2     The solar system consists of the Sun, eight planets, their moons, dwarf planets, and
3     The inner planets—Mercury, Venus, Earth, and Mars—are rocky and solid.
4     The outer planets—Jupiter, Saturn, Uranus, and Neptune—are much larger and gaseous.
5 """
6 question = "Which planets in the solar system are rocky and solid?"
7 template = """
8     Answer the {question} based on the {content}.
9     Respond "Unsure about answer" if not sure about the answer.
10    Answer:
11 """
12 prompt = PromptTemplate.from_template(template)
13 output_key = "answer"
14 llm_chain = LLMChain(prompt=prompt, llm=mixtral_llm, output_key=output_key)
15 response = llm_chain.invoke(input = {"question":question, "content": content})
16 print(response["answer"])

```

Code generation

An agent that is designed to generate SQL queries based on given descriptions. It interprets the requirements from your input and translates them into executable SQL code.

Code Example:

```

1 description = """
2     Retrieve the names and email addresses of all customers from the 'customers' table
3     The table 'purchases' contains a column 'purchase_date'
4 """
5 template = """
6     Generate an SQL query based on the {description}
7     SQL Query:
8 """
9 prompt = PromptTemplate.from_template(template)
10 output_key = "query"
11 llm_chain = LLMChain(prompt=prompt, llm=mixtral_llm, output_key=output_key)
12 response = llm_chain.invoke(input = {"description":description})
13 print(response["query"])

```

Role playing

Configures the LLM to assume specific roles as defined by us, enabling it to follow predetermined rules and behave like a task-oriented chatbot.

Code Example:

```

1 role = """
2     game master
3 """
4 tone = "engaging and immersive"
5 template = """
6     You are an expert {role}. I have this question {question}. I would like our con
7     Answer:
8 """
9 prompt = PromptTemplate.from_template(template)
10 output_key = "answer"
11 llm_chain = LLMChain(prompt=prompt, llm=mixtral_llm, output_key=output_key)

```

class_names

This code snippet maps numerical labels to their corresponding textual descriptions to classify tasks. This code helps in machine learning to interpret the output model, where the model's predictions are numerical and should be presented in a more human-readable format.

Code Example:

```

1 class_names = {0: "negative", 1: "positive"}
2 class_names

```

read_and_split_text

Involves opening the file, reading its contents, and splitting the text into individual paragraphs. Each paragraph represents a section of the company policies. You can also filter out any empty paragraphs to clean your data set.

Code Example:

```

1 def read_and_split_text(filename):
2     with open(filename, 'r', encoding='utf-8') as file:
3         text = file.read()
4         # Split the text into paragraphs (simple split by newline characters)
5         paragraphs = text.split('\n')
6         # Filter out any empty paragraphs or undesired entries
7         paragraphs = [para.strip() for para in paragraphs if len(para.strip()) > 0]
8         return paragraphs
9     # Read the text file and split it into paragraphs
10    paragraphs = read_and_split_text('companyPolicies.txt')
11    paragraphs[0:10]

```

encode_contexts

This code snippet encodes a list of texts into embeddings using `content_tokenizer` and `context_encoder`. This code helps iterate through each text in the input list, tokenizes and encodes it, and then appends the pooler_output to the embeddings list. The resulting embeddings get stored in the `context_embeddings` variables and generate embeddings from text data for various natural language processing (NLP) applications.

Code Example:

```

1 def encode_contexts(text_list):
2     # Encode a list of texts into embeddings
3     embeddings = []
4     for text in text_list:
5         inputs = context_tokenizer(text, return_tensors='pt', padding=True, truncation=True)
6         outputs = context_encoder(**inputs)
7         embeddings.append(outputs.pooler_output)
8     return torch.cat(embeddings).detach().numpy()
9     # you would now encode these paragraphs to create embeddings.
10    context_embeddings = encode_contexts(paragraphs)

```

import faiss

FAISS (Facebook AI Similarity Search) is an efficient library developed by Facebook for similarity search and clustering of dense vectors. FAISS is designed for fast similarity search, which is particularly valuable when dealing with large data sets. It is highly suitable for tasks in natural language processing where retrieval speed is critical. It effectively handles large volumes of data, maintaining performance even as data set sizes increase.

Code Example:

```

1 import faiss
2 # Convert list of numpy arrays into a single numpy array
3 embedding_dim = 768 # This should match the dimension of your embeddings
4 context_embeddings_np = np.array(context_embeddings).astype('float32')
5 # Create a FAISS index for the embeddings
6 index = faiss.IndexFlatL2(embedding_dim)
7 index.add(context_embeddings_np) # Add the context embeddings to the index

```

search_relevant_contexts

This code snippet is useful in searching relevant contexts for a given question. It tokenizes the question using the `question_tokenizer`, encodes the question using `question_encoder`, and searches an index for retrieving the relevant context based on question embedding.

Code Example:

```

1 def search_relevant_contexts(question, question_tokenizer, question_encoder, index, k=5):
2     """
3     Searches for the most relevant contexts to a given question.
4     Returns:
5     tuple: Distances and indices of the top k relevant contexts.
6     """
7     # Tokenize the question
8     question_inputs = question_tokenizer(question, return_tensors='pt')
9     # Encode the question to get the embedding
10    question_embedding = question_encoder(**question_inputs).pooler_output.detach().numpy()
11    # Search the index to retrieve top k relevant contexts
12    D, I = index.search(question_embedding, k)
13    return D, I

```

generate_answer_without_context

This code snippet generates responses using the entered prompt without requiring additional context. It tokenizes the input questions using the tokenizer, generates the output text using the model, and decodes the generated text to obtain the answer.

Code Example:

```
1 def generate_answer_without_context(question):
2     # Tokenize the input question
3     inputs = tokenizer(question, return_tensors='pt', max_length=1024, truncation=True)
4     # Generate output directly from the question without additional context
5     summary_ids = model.generate(inputs['input_ids'], max_length=150, min_length=40, length_penalty=2.0, num_beams=4)
6     # Decode and return the generated text
7     answer = tokenizer.decode(summary_ids[0], skip_special_tokens=True)
8     return answer
```

Generating answers with DPR contexts

Answers are generated when the model utilizes contexts retrieved via DPR, which are expected to enhance the answer's relevance and depth.

Code Example:

```
1 def generate_answer(contexts):
2     # Concatenate the retrieved contexts to form the input to BART
3     input_text = ' '.join(contexts)
4     inputs = tokenizer(input_text, return_tensors='pt', max_length=1024, truncation=True)
5     # Generate output using BART
6     summary_ids = model.generate(inputs['input_ids'], max_length=150, min_length=40, length_penalty=2.0, num_beams=4)
7     return tokenizer.decode(summary_ids[0], skip_special_tokens=True)
```

aggregate_embeddings function

The function `aggregate_embeddings` takes token indices and their corresponding attention masks, and uses a BERT model to convert these tokens into word embeddings. It then filters out the embeddings for zero-padded tokens and computes the mean embedding for each sequence. This helps in reducing the dimensionality of the data while retaining the most important information from the embeddings.

Code Example:

```
1 def aggregate_embeddings(input_ids, attention_masks, bert_model=bert_model):
2     """
3     Converts token indices and masks to word embeddings, filters out zero-padded embeddings,
4     and aggregates them by computing the mean embedding for each input sequence.
5     """
6     mean_embeddings = []
7     # Process each sequence in the batch
8     print('number of inputs', len(input_ids))
9     for input_id, mask in tqdm(zip(input_ids, attention_masks)):
10         input_ids_tensor = torch.tensor([input_id]).to(DEVICE)
11         mask_tensor = torch.tensor([mask]).to(DEVICE)
12         with torch.no_grad():
13             # Obtain the word embeddings from the BERT model
14             word_embeddings = bert_model(input_ids_tensor, attention_mask=mask_tensor)[0].squeeze(0)
15             # Filter out the embeddings at positions where the mask is zero
16             valid_embeddings_mask = mask_tensor[0] != 0
17             valid_embeddings = word_embeddings[valid_embeddings_mask, :]
18             # Compute the mean of the filtered embeddings
19             mean_embedding = valid_embeddings.mean(dim=0)
20             mean_embeddings.append(mean_embedding)
21     # Concatenate the mean embeddings from all sequences in the batch
22     aggregated_mean_embeddings = torch.cat(mean_embeddings)
```

text_to_emb

Designed to convert a list of text strings into their corresponding embeddings using a pre-defined tokenizer.

Code Example:

```
1 def text_to_emb(list_of_text, max_input=512):
2     data_token_index = tokenizer.batch_encode_plus(list_of_text, add_special_tokens=True, padding=True, max_length=max_input)
3     return question_embeddings
```

process_song

Convert both the predefined appropriateness questions and the song lyrics into "RAG embeddings" and measure the similarity between them to determine the appropriateness.

Code Example:

```
1 import re
2 def process_song(song):
```

```

1 def process_song(song):
2     # Remove line breaks from the song
3     song_new = re.sub(r'\n', ' ', song)
4     # Remove single quotes from the song
5     song_new = [song_new.replace("'", "")]
6     return song_new

```

RAG_QA

This code snippet performs question-answering using question embeddings and provides embeddings. It helps reshape the results for processing, sorting the indices in descending order, and printing the top 'n-responses' based on the highest dot product values.

Code Example:

```

1 def RAG_QA(embeddings_questions, embeddings, n_responses=3):
2     # Calculate the dot product between the question embeddings and the provided embeddings
3     dot_product = embeddings_questions @ embeddings.T
4     # Reshape the dot product results to a 1D tensor for easier processing.
5     dot_product = dot_product.reshape(-1)
6     # Sort the indices of the dot product results in descending order (setting descending to True)
7     sorted_indices = torch.argsort(dot_product, descending=True)
8     # Convert sorted indices to a list for easier iteration.
9     sorted_indices = sorted_indices.tolist()
10    # Print the top 'n_responses' responses from the sorted list, which correspond to the highest dot product values.
11    for index in sorted_indices[:n_responses]:
12        print(embeddings[index])

```

model_name_or_path

This code snippet defines the model name to 'gpt2' and initializes the token and model using the GPT-2 model. In this code, add special tokens for padding by keeping the maximum sequence length to 1024.

Code Example:

```

1 # Define the model name or path
2 model_name_or_path = "gpt2"
3 # Initialize tokenizer and model
4 tokenizer = GPT2Tokenizer.from_pretrained(model_name_or_path, use_fast=True)
5 model = GPT2ForSequenceClassification.from_pretrained(model_name_or_path, num_labels=1)
6 # Add special tokens if necessary
7 tokenizer.pad_token = tokenizer.eos_token
8 model.config.pad_token_id = model.config.eos_token_id
9 # Define the maximum length
10 max_length = 1024

```

add_combined_columns

This code snippet combines the prompt with chosen and rejected responses in a data set example. It combines with the 'Human:' and 'Assistant:' for clarity. This function modifies each example in the 'train' split of the data set by creating new columns 'prompt_chosen' and 'prompt_rejected' with the combined text.

Code Example:

```

1 # Define a function to combine 'prompt' with 'chosen' and 'rejected' responses
2 def add_combined_columns(example):
3     # Combine 'prompt' with 'chosen' response, formatting it with "Human:" and "Assistant:"
4     example['prompt_chosen'] = "\n\nHuman: " + example["prompt"] + "\n\nAssistant: " + example["chosen"]
5     # Combine 'prompt' with 'rejected' response, formatting it with "Human:" and "Assistant:"
6     example['prompt_rejected'] = "\n\nHuman: " + example["prompt"] + "\n\nAssistant: " + example["rejected"]
7     # Return the modified example
8     return example
9 # Apply the function to each example in the 'train' split of the dataset
10 dataset['train'] = dataset['train'].map(add_combined_columns)

```

RetrievalQA

This code snippet creates an example for 'RetrievalQA' using a language model and document retriever.

Code Example:

```

1 qa = RetrievalQA.from_chain_type(llm=flan_ul2_llm,
2                                 chain_type="stuff",
3                                 retriever=docs_retriever)
4 query = "what is mobile policy?"
5 qa.invoke(query)

```

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